EXPERIMENT 6

Read an image and perform erosion using the Erode function.

Aim

Read an image and perform erosion using the Erode function.

Algorithm

• Import OpenCV and NumPy.

• Read the image and convert it to grayscale.

• Convert the grayscale image to binary form.

• Create a structuring kernel and apply erosion.

• Save the eroded image.

Code

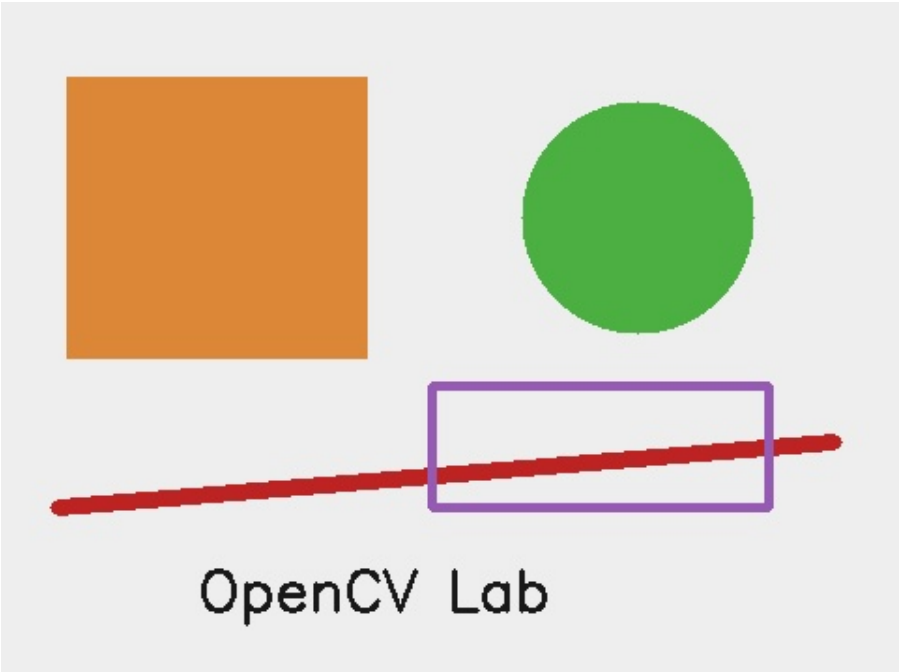
```
import cv2
import numpy as np

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
_, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
result = cv2.erode(binary, np.ones((5, 5), np.uint8), iterations=1)
cv2.imwrite("output_images/experiment06_erosion.png", result)
print("Saved output_images/experiment06_erosion.png")
```

Input Image:

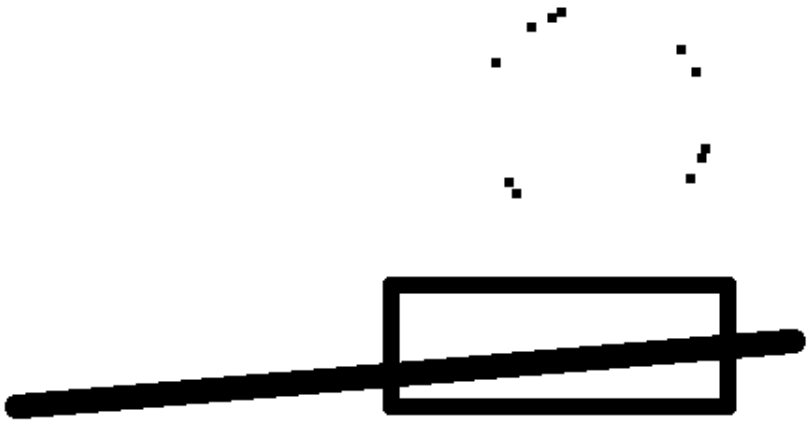
Output Image:

Note: Morphological operations use a thresholded binary version of the sample image.

Input Image



Output Image



OpenCV Lab